Course Code: MTH2802	**Course Name:** Linear Algebra & Differential Equations	
Credits: 3 (3-0-0)	**Contact hours per week:** 3 hours of lecture per week	
Batch: 2024, 3rd Sem **Academic Year:** 2025-26	**Semester Duration:** 29 July – 28 Nov. 2025	
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Course Overview:

This course offers a comprehensive introduction to the essential principles of Linear Algebra and Ordinary Differential Equations (ODEs), with a strong focus on their applications in engineering, computer science, and data-driven domains. Students will explore the structure and behavior of linear systems, vector spaces, eigenvalue problems, and linear transformations, alongside various methods for solving differential equations that arise in real-world models. By the end of the course, students will be able to translate real-world problems into mathematical frameworks, solve them using appropriate methods, and interpret results effectively for practical use.

Salient topics of the course are listed below:

Systems of Linear Equations:

Row echelon form (REF) and reduced-row echelon form (RREF) of a matrix, Rank of a matrix, Linear equations, and their solutions using Gaussian elimination method and Gauss-Jordan method, Applications of Systems of Linear equations.

Eigenvalues and Eigenvectors:

Introduction to eigenvalues and eigenvectors, its applications in Markov chain, dynamical systems and system of differential equations, diagonalization of matrices.

Vector Spaces:

Introduction to vector spaces, subspaces, linear combination of vectors, spanning set of vectors, linear dependence and independence of vectors, basis and dimension of a vector space, Linear transformation and its matrix representation, Kernel and range of Linear transformation.

First Order Differential Equations:

Solution of Exact, linear and Bernoulli's differential equations, applications of differential equations of first order & first degree to simple electric circuits, Newton's law of cooling and orthogonal trajectories.

Higher Order Differential Equations:

Understanding the reduction of orders, Solution of higher order linear differential equations with constant coefficients, Cauchy Euler equation, Method of undetermined coefficients, Variation of Parameters, Solution of Cauchy's and Legendre's equations, Modelling and solving linear differential equations of 2nd and higher orders in engineering, solution of system of linear differential Equations.

Week-Wise Schedule:

Week 1: REF and RREF of a matrix, Rank of a matrix, Introduction to linear equations.

Week 2-3: Solution of linear system of equations using Gaussian elimination method and Gauss-Jordan method, Applications of Systems of Linear equations.

Week 4-5: Introduction to eigenvalues and eigenvectors, its applications in Markov chain, dynamical systems and system of differential equations, diagonalization of matrices.

Week 6-8: Introduction to vector spaces, subspaces, Linear dependence and independence of vectors, Linear combination of vectors, spanning set of vectors, basis and dimension of a vector space, Linear transformation and its matrix representation, Kernel and range of Linear transformation.

Week 9-10: Solution of Exact, linear and Bernoulli's differential equations, applications of differential equations of first order & first degree to simple electric circuits, Newton's law of cooling and orthogonal trajectories.

Week 11-13: Understanding the reduction of orders, Solution of higher order linear differential equations with constant coefficients, Method of undetermined coefficients, Variation of Parameters.

Week 14-16: Solution of Cauchy-Euler and Legendre's equations, Modelling and solving linear differential equations of 2nd and higher orders in engineering, solution of system of linear differential Equations.

Course Outcomes (COs):

	Course Outcome Statement	Mapped Component	Associated Activities/Assessment
CO1	Understand various concepts of linear systems, vector spaces, eigenvalues and eigenvectors, linear transformation, different types of ordinary differential equations that arise in different fields of engineering problems.	Skill Development	Quizzes, assignments, end-term exam
CO2	Apply the concepts of linear systems, eigenvalue, eigenvectors, linear transformations in graphics and other engineering domains, apply suitable methods to solve different types of ODEs arising in engineering and real-world applications.	Skill Development	Quizzes, assignments, end-term examination
CO3	Analyze the long-term behavior of Markov chains and discrete/continuous dynamical systems, analyze the efficiency of different solution methods available to solve higher order ODEs.	Skill Development	Assignments, end-term examination, project
CO4	Model real world problems and finding their solutions mathematically as well as with the help of technology and writing a report to summarize the findings.	Skill Development, Employability	Project work, report writing, presentations

CO-PO Mapping for CSE:

PO and PSO →	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**	**PSO3**
↓ CO Mapping || Course Mapping →															
CO1	1														
CO2	3			1	1										
CO3	3			1	1										
CO4	3			2	1					2	2				

For details about PO, PSO, and CO-PO mapping, refer to the OBE document for B. Tech. – CSE program.

1-**Low Correlation; 2- Moderate Correlation; 3- Substantial Correlation**

CO-PO Mapping for ME:

PO and PSO →	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO1	PSO2	PSO3
↓ CO Mapping || Course Mapping →															
CO1	1														
CO2	3			1	1										
CO3	3			1	1										
CO4	3			2	1					2	2				

For details about PO, PSO, and CO-PO mapping, refer to the OBE document for B. Tech. –ME program.

1-**Low Correlation; 2- Moderate Correlation; 3- Substantial Correlation**

CO-PO Mapping for EComE:

PO and PSO →	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO1	PSO2	PSO3
↓ CO Mapping || Course Mapping →															
CO1	1														
CO2	3			1	1										
CO3	3			1	1										
CO4	3			2	1					2	2				

For details about PO, PSO, and CO-PO mapping, refer to the OBE document for B. Tech. – EComE program.

1-**Low Correlation; 2- Moderate Correlation; 3- Substantial Correlation**

Course Competencies (Course Outcomes further elaborated) and Instruction schedule:

	Competency	**CO**	**CL**	**# Lecture Sessions**
C1	● Understand the concept matrices and elementary row operations. ● Apply elementary row operations to matrix rank related problems.	CO1, CO2	U AP	3
C2	● Understand and use Gauss Elimination and Gauss Jordan Methods for solving linear systems of equations. ● Understand and compute eigen values and eigen vectors of square matrices. ● Analyze the pitfalls and advantages of using Gauss Elimination and Gauss Jordan Methods for solving linear systems of equations. ● Apply the knowledge of eigen values and eigen vectors in dynamical systems and matrix diagonalization.	CO1, CO2, CO3	U AP AN	12
C3	● Understand the concepts of differential equations of first order. ● Apply the knowledge of first order differential equations for simple electric circuits and Newton's law of cooling. ● Evaluate the solution of exact and Bernoulli's differential equations.	CO1, CO2, CO3, CO4	U AP AN EV	9
C4	● Understand the concept of reduction of order of differential equations. ● Apply the method of complementary functions and particular integral to solve higher order differential equations with constant coefficients. ● Evaluate the solution of higher order ODEs using the method of undetermined coefficients and the method of variation of parameters.	CO1, CO2, CO3, CO4	U AP AN EV	12
C5	● Compute the solution of Cauchy's and Legendre's differential equations ● Model real world engineering problems using ordinary differential equations.	CO3, CO4	AP EV	12
	Total contact hours=48			

Learning Resources:

Textbooks (TB)

1. D. Poole, *Linear Algebra: A Modern Introduction*, 4th Edition, ISBN-13: 978-1-285-46324-7, Cengage Learning, 2015.
2. Zill D. G., *A First Course in Differential Equations with Modelling Applications*, 11th Edition. ISBN-13: 978-0-495-10824-5, Brooks/Cole, Cengage Learning, 2010.

Reference Books (RB)

- David C. Lay, *Linear Algebra and Its Applications*, ISBN: 978-81-775-8333-5, 3rd Edition, Pearson, 2014.
- Seymour Lipschutz & Marc Lipson, *Schaum's Outline of Linear Algebra*, ISBN: 978-0071794565, 5th Edition, McGraw-Hill, 2014.
- Simmons G.F, *Differential Equations with Applications and Historical Notes*, ISBN: 13:978-1-4987-0259-1, 3rd edition, CRC press, Taylor & Francis Group, 2017.

Note: Instructor will regularly post the necessary learning resources such as lecture resources, information related to assignments etc., to the online course management portal i.e. Maitri/Google Classroom. Additional resources related to open access software etc. will be provided on a timely basis.

Assessment Pattern: The final grade will be based on the marks/ grades obtained in the mid-semester and end-semester evaluation along with other assessments defined in the assessment table. Relative grading methods defined in the academic regulations of the university will be followed to grade the students.

Evaluation Component	Weightage (%)	Evaluation Schedule	Rubrics/Remarks
Quiz	20 (2 quizzes each of weightage 10%)	August 25, 2025 October 9, 2025	Quiz is a 25-minute-long closed book assessment component. Understanding of Concepts (40%) • Proper understanding of concepts of Linear algebra and differential equation topics. • Validation of students' understanding through timed quizzes. Problem Solving (60%) • Applying the knowledge to solve various engineering problems • Logical ability to problem-solving skills in a timely manner.

Assignment	20 (2 assignment each of weightage 10%)	September 15, 2025 October 28, 2025	Open Book assignment of weight 10% followed by a closed book test of another 10%. Understanding of Concepts (20%) ● Detailed understanding of various concepts and terminologies of the subject. ● Understanding the importance of data in decision making. ● Basic time management skill through a thorough timed test. Timeliness (20%) ● Assignment must be submitted on or before the due date and time. Problem Solving (60%) ● Applying mathematical knowledge to solve various problems arising in different engineering domains. ● Developing problem-solving and time management skills.
Project	20	November 18, 2025 (Presentation: Tentatively during Nov 26-28, 2025, in the class timings during revision week)	Understanding of Concepts (10%) ● Comprehensive understanding of the concepts taught throughout the course. ● Understanding of various types of data and their basic characteristics. Timeliness (10%) ● Report must be submitted on or before the due date and time. Problem Solving & Report writing (30%) ● Applying graphical tools and software to present the findings visually for better understanding. ● Developing in-depth understanding to prepare a detailed report of the findings. Presentation & Viva (50%)

			● Developing peer-learning and collaboration skills. ● Developing presentation skills to present the work in a group. ● Demonstration of individual knowledge through viva.
End Semester	40	As per University Calendar	Comprehensive Knowledge (40%) ● Demonstrates understanding of the entire course material. ● Ability to apply concepts to solve complex problems. Practical Application (40%) ● Application of theoretical knowledge in practical scenarios. Critical Thinking (20%) ● Analytical skills and innovative solutions to handle real-life problems. • Ability to analyze the efficiency of various available methods to apply the suitable one for time optimization and improve credibility

Experiential Learning Components:

To reinforce theoretical knowledge through practical application, this course integrates experiential learning via hands-on assignments and project. These components are designed to strengthen problem-solving and modelling skills in engineering and data driven contexts. Students will:

- Formulate mathematical models by translating real world for industry inspect problems using principles from linear algebra and ordinary differential equations.
- Use Python, Matlab, Mathematica or any other as a computational tool to solve and analyze these models' incorporating libraries for numerical computing, symbolic algebra and data visualization.
- Interpret and visualize results using plotting tools to understand system behavior and support decision making.
- Preparing detailed project reports that summarize findings and insights.
- Presenting project outcomes to demonstrate practical applications of the learned concepts.

Student Responsibilities:

• Attend lectures to obtain all the course material that you are responsible for.

• Check announcements at Maitri/Google Classroom on a regular basis.

• Submit assignments/projects on time.

Regularly, check your marks on the Maitri and inform the instructor in-charge in case of any issues.

Attendance Policy: Students are expected to attend the classes regularly. Failure to attend the classes regularly and adhere to the expected attendance percentage may result in a reduction of the grade or awarding 'R' grade as per the University's grading policy.

Recourse examination policy: As per institute policy.

Make-up examination/work policy: As per University policies.

Behavior expectations: No mobile phones and other distractive elements and gadgets in the class.

Other Policies or Procedures:

a) Academic Honesty

Students must abide by BMU's academic honesty policy. For example, an excerpt is provided for guidance as follows:

"Students who compromise the integrity of the classroom are subject to disciplinary action on the part of the University. Violations of classroom standard include:

1. Cheating in any form, whether in formal examinations or elsewhere

2. Plagiarism, using the work of others as your own without assigning proper credit to the source.

3. Misrepresentation of any work done in the classroom or in preparation for class.

4. Disruptive behavior in a course of study or abusiveness toward faculty or fellow students."

b) Teaching Improvement

Please feel free to make suggestions to improve the content of the class and my instruction skills. You can share these suggestions directly with me and/or via email. You can also leave your comments anonymously by sending online messages.